

Probability – 4

The probabilistic method across mathematics

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Theoretical minimum.

Theorem 8 (the two forms). If $\mathbb{E} X \geq c$, then $X \geq c$ for some outcome; if $\Pr(\text{bad}) < 1$, then a good outcome exists. Nothing more is needed for most applications: one invents a probability distribution on the objects in question and computes one expectation.

Theorem 9 (union bound, alteration, local lemma). $\Pr(\bigcup A_i) \leq \sum \Pr(A_i)$.

If this is not enough, one takes a random object and *repairs* it by deleting the few bad parts — the alteration method. If the bad events are sparsely dependent, the Lovász local lemma still gives $\Pr(\bigcap \overline{A_i}) > 0$; the union bound and the symmetric local lemma were proved in Inequalities – 8.

Fact 4 (where it works outside combinatorics). In linear algebra: a random vector avoids any fixed proper subspace with probability one, so a generic matrix has full rank and a property that holds with positive probability holds for some matrix. In number theory: a density statement about integers is a probability statement about a uniform residue, and equidistribution turns Weyl sums into expectations. In analysis: Bernstein polynomials are expectations of f at a binomial random point, and the law of large numbers proves the Weierstrass theorem.

Fact 5 (second moment and entropy). If $\mathbb{E} X > 0$ and $\mathbb{E} X^2 \leq C(\mathbb{E} X)^2$, then $\Pr(X > 0) \geq 1/C$, which is how existence with high probability is upgraded to a positive proportion. Shearer’s entropy inequality bounds the entropy of a random object by the entropies of its projections and is the probabilistic form of the Loomis–Whitney inequality.

1. Prove Theorem 8 and the alteration method, and use them for the two classical examples: $R(k, k) > 2^{k/2}$, and every graph has a bipartite subgraph containing at least half of its edges.
2. (Putnam 2004) An $m \times n$ checkerboard is coloured at random, each square independently red or black with probability $\frac{1}{2}$. Two squares are in the same monochromatic region if they are joined by a path of squares of one colour through common sides. Show that the expected number of monochromatic regions exceeds $\frac{mn}{8}$.
3. (Iranian competition) Prove that no set of nilpotent matrices spans $M_n(\mathbb{F})$, and give the argument in the form “a random matrix has non-zero trace”.
4. (IMC 2025 shortlist) Fix N and let M_N be the 2×2 matrices with entries in $\{1, \dots, N\}$. Two matrices A, B are chosen independently and uniformly.
 - a) Show that the number of commuting ordered pairs is $O(N^6)$.
 - b) Deduce that $\Pr(AB = BA) = O(N^{-2})$, so the probability tends to zero.
5. (Putnam 2016) Let A be a $2n \times 2n$ matrix whose entries are independent and equal to 0 or 1 with probability $\frac{1}{2}$ each. Find the expected value of $\det(A - A^T)$.